CHAPTER 1

INTRODUCTION

Communication is a fundamental aspect of human interaction. For the Deaf and Hard-of-Hearing (DHH) community, sign language serves as the primary medium of communication. However, the lack of widespread understanding of sign language among the general population creates a significant communication barrier. Although advancements in artificial intelligence and computer vision have led to the development of automated sign language recognition systems, most existing solutions focus on global or widely used languages such as American Sign Language (ASL) or Indian Sign Language (ISL). Regional sign languages like Kannada Sign Language (KSL) remain largely unexplored and unsupported by robust technological systems. This project proposes a Deep Learning and NLP-based real-time sign language recognition system that focuses on Kannada Vyanjanagalu (consonants). The system captures live hand gestures using a webcam, recognizes them using a trained Convolutional Neural Network (CNN), constructs meaningful Kannada text using rule-based Natural Language Processing (NLP), and finally converts the text into natural Kannada speech. The objective is to provide an end-to-end, real-time, and context-aware communication solution that enhances accessibility and inclusion for the DHH community.

1.1 EXISTING SYSTEM

Existing sign language recognition systems primarily focus on widely adopted sign languages such as ASL and ISL. These systems generally rely on traditional machine learning techniques or basic deep learning models to classify static or dynamic hand gestures. Many approaches require specialized hardware such as data gloves, motion sensors, or depth cameras, which increases cost and limits real-world usability. Some recent systems utilize computer vision techniques with deep learning models but lack contextual understanding. Furthermore, most existing solutions do not support regional languages like Kannada, nor do they address the complexity of regional scripts involving vowel attachment and conjunct consonants (vattakshara). Another major limitation of current systems is the absence of real-time performance and natural speech output. Systems that do offer speech synthesis frequently suffer from latency, inaccurate pronunciation, or lack of support for regional languages. As a result, existing systems are insufficient for seamless, real-time communication in practical scenarios.

1.2 PROPOSED SYSTEM

The proposed system introduces a real-time, web-based Kannada Sign Language recognition platform using deep learning and NLP techniques. The system captures live gesture input through a standard webcam and processes it using a CNN trained specifically on Kannada Vyanjanagalu gestures. The CNN extracts spatial features such as finger orientation, hand shape, and palm structure to accurately classify gestures. Unlike existing systems, the proposed approach integrates a rule-based NLP module to construct meaningful Kannada text. This module applies linguistic rules such as vowel attachment, vattakshara formation, duplicate suppression, and automatic spacing to generate grammatically correct Kannada words. The system also includes a hybrid text-to-speech mechanism, utilizing the browser-based Web Speech API with a fallback gTTS engine to generate clear and natural Kannada speech. The entire system is deployed as a web application using modern frameworks, ensuring real-time performance, low latency, and ease of accessibility without requiring specialized hardware.

1.3 OBJECTIVES

The main objectives of the project are as follows:

- To design and train a CNN-based deep learning model for accurate recognition of Kannada Vyanjanagalu hand gestures.
- To develop a rule-based NLP module for constructing meaningful and contextually correct Kannada text.
- To implement a real-time sign-to-text and sign-to-speech conversion system.
- To ensure low-latency performance suitable for real-time applications.
- To build a web-based, user-friendly interface that works with a standard webcam.

1.4 PROBLEM STATEMENT

Despite advancements in sign language recognition technologies, there is a lack of effective and real-time solutions for Kannada Sign Language. Existing systems either focus on non-regional sign languages, require expensive hardware, or fail to generate contextually meaningful text. Additionally, Kannada script requires complex linguistic rules for word formation, making direct gesture-to-text mapping insufficient. Therefore, there is a need for an AI-driven, real-time, low-latency system that can accurately recognize Kannada sign gestures, generate grammatically correct Kannada text, and convert it into natural speech to facilitate smooth communication between the DHH community and the general population.

1.5 HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements

- Webcam (built-in or external)
- ARCore not required; standard RGB camera sufficient
- Development System: Processor: Intel i3/i5 or higher, RAM: Minimum 8 GB (16 GB recommended), Storage: Minimum 20 GB free space
- GPU (optional): For faster model training

Software Requirements

- Operating System: Windows 10 / Windows 11
- Programming Language: Python 3.10
- Deep Learning Frameworks: TensorFlow, Keras
- Computer Vision Library: OpenCV
- NLP Tools: Rule-based Kannada grammar module
- Web Framework: FastAPI
- Frontend: HTML, CSS, JavaScript, React.js
- Speech Synthesis: Web Speech API, gTTS
- Development Tools: Google Colab, VS Code

1.6 SCOPE OF THE PROJECT

The scope of this project includes the development of a real-time Kannada Sign Language recognition system capable of recognizing consonant gestures and converting them into text and speech. The system is designed to work under real-world conditions with variations in lighting and background.

The project can be extended to support:

- Kannada vowels (Swaragalu)
- Compound and continuous gestures
- Sentence-level recognition
- Mobile application deployment
- MediaPipe-based landmark tracking
- Integration with educational and assistive platforms

The system has potential applications in education, healthcare, public services, and customer support environments.

CHAPTER 2

LITERATURE SURVEY

There are different techniques that have been explored for sign language identification and translation, ranging from classical machine learning methods to modern deep learning architectures. These techniques can broadly be categorized into three key domains:

- Gesture Recognition Algorithms: Early systems relied on hybrid approaches combining handcrafted feature extraction with classifiers such as Support Vector Machines (SVMs) and Hidden Markov Models (HMMs). However, these methods struggled with illumination changes, gesture complexity, and scalability. In contrast, Convolutional Neural Networks (CNNs) have demonstrated remarkable accuracy in visual feature extraction and classification, especially for static sign alphabets like those in American Sign Language (ASL). CNN-based models automatically learn spatial hierarchies and are computationally efficient for real-time hand gesture recognition tasks.
- Real-Time Processing Methods: To improve recognition speed and robustness, modern frameworks like MediaPipe and OpenPose enable efficient hand keypoint detection and skeletal tracking. These systems reduce dependency on raw pixel data and provide structural cues that improve real-time accuracy. While Transformer-based architectures and attention mechanisms have also been explored for continuous sign language recognition, CNNs remain more practical for alphabet-level real-time inference, especially on edge and browser-based systems.
- NLP and Sentence Generation Models: Once the alphabet or gesture sequence is recognized, Natural Language Processing (NLP) techniques are applied to construct coherent sentences and generate speech. Lightweight models and APIs such as Google Text-to-Speech (gTTS) or the Web Speech API are commonly used to convert recognized text into audible speech. Although advanced models such as BERT, GPT, and Neural Machine Translation (NMT) are capable of contextual sentence generation, simpler rule-based NLP methods are sufficient for alphabet-level ASL recognition systems, ensuring fast and accurate real-time translation.

Boobal et al. (2024) developed a real-time sign language and audio conversion system that integrates Convolutional Neural Networks (CNNs) and Natural Language Processing (NLP) to facilitate two-way communication for hearing-impaired and non-signing individuals. Their model employs MobileNet for gesture classification and Google's Speech API for audio-to-sign translation, enabling seamless conversion between spoken language and animated sign gestures. The system architecture supports Indian Sign Language (ISL) and emphasizes real-time responsiveness, user comfort, and multilingual adaptability. However, challenges such as gesture ambiguity, speech variability, and dataset limitations persist, necessitating further refinement in gesture mapping and contextual understanding. [1]

Bayrak et al. (2024) introduced a novel American Sign Language (ASL) recognition model using Complex Zernike Moments (CZMs) for features extraction and Complex-Valued Deep Neural Networks (CVDNNs) for classification. Unlike traditional real-valued models, their approach leverages complex numbers to capture both magnitude and phase information, enhancing recognition accuracy and robustness. The model achieved 89.01% accuracy on the Sign Language MNIST dataset and up to 98.67% on the Massey University dataset using holdout validation. This study demonstrates the potential of the complex-valued architectures in improving precision and computational efficiency in sign language recognition, especially for static gestures. [2]

Mukesh Kumar et al. (2024) proposed an LSTM-driven sign language recognition system that translates gestures into speech using MediaPipe Holistic for landmark detection and Pyttsx3 for text-to-speech synthesis. Their architecture captures temporal dependencies in dynamic gestures, achieving 91.63% accuracy across a custom ASL dataset. The system supports real-time gesture recognition and audio output, bridging communication gaps for the deaf community. Despite its effectiveness, the model faces challenges in recognizing nuanced facial expressions and multi-modal signs, indicating a need for expanded datasets and multimodal fusion techniques. [3]

Venkadesh et al. (2024) presented SignChatAI, a generative AI-based communication tool tailored for the deaf-and-mute community using Indian Sign Language. Their system combines CNNs for image processing with LSTM networks to capture sequential gesture patterns. The model supports bidirectional translation between ISL and text, with accuracy levels ranging from 90% to 100%. SignChatAI emphasizes inclusivity and accessibility, offering a better scalable solution for gesture-based communication. However, the system's reliance on predefined datasets and limited gesture

vocabulary highlighting the importance of continuous learning and dataset expansion to accommodate regional variations and complex expressions. [4]

Soundarya M et al. (2024) proposed a machine learning–based sign language recognition system aimed at improving communication between hearing and hearing-impaired individuals. The system primarily focuses on American Sign Language (ASL) alphabet recognition and employs Long Short-Term Memory (LSTM) networks to effectively capture temporal dependencies in sequential gesture data. Unlike conventional CNN-based image-only approaches, this work integrates LSTM models to process continuous video streams, enabling better understanding of dynamic hand movements. OpenCV is used for real-time data capture, while TensorFlow handles preprocessing, feature extraction, and model training. The proposed system achieves accuracy levels ranging from 90% to 100%, with an average recognition accuracy of 98.13%, demonstrating its effectiveness in recognizing complex sign patterns. [5]

Kothadiya D et al. (2023) proposed an explainable artificial intelligence (XAI)–enabled framework named SignExplainer for improving transparency and trust in sign language recognition systems. The study introduces an attention-based ensemble learning architecture that combines ResNet-50 with a self-attention mechanism to effectively learn discriminative spatial features from static sign language images. The ensemble model employs a bagging strategy and global feature embedding to enhance recognition performance, achieving a high accuracy of 98.20% on Indian Sign Language (ISL) datasets, along with strong results on American and Bangla Sign Language datasets. To address the black-box nature of deep learning models, the authors integrate SHAP-based explainability techniques to interpret prediction decisions by highlighting relevant hand regions contributing to classification outcomes. The proposed SignExplainer framework provides visual and quantitative explanations, improving user trust and model interpretability. Experimental results demonstrate superior performance compared to conventional CNN, VGG16, EfficientNet, and traditional machine learning models. However, the framework is limited to static sign recognition and requires high computational resources, suggesting future extensions toward real-time dynamic gesture recognition, multimodal inputs, and lightweight explainable models for practical deployment. [6]

Papastratis et al. (2020) proposed a novel continuous sign language recognition (CSLR) framework that leverages cross-modal learning by aligning video and text embeddings into a shared joint latent

space. The system employs a CNN-based video encoder combined with temporal convolution layers and bidirectional LSTM networks to capture spatiotemporal dependencies in sign language videos. In parallel, a text encoder based on LSTM models the linguistic structure of gloss sequences. Both video and text representations are mapped into a common latent space using alignment loss functions, enabling effective modeling of intra-gloss dependencies. The aligned video embeddings are decoded using a jointly trained decoder with Connectionist Temporal Classification (CTC) loss. Experimental evaluation on benchmark datasets such as RWTH-Phoenix-Weather-2014 demonstrated significant improvements in recognition accuracy compared to conventional vision-only approaches. The key contribution of this work lies in exploiting textual information during training to enhance visual representations. However, the approach involves high computational complexity and requires large annotated datasets, which may limit its scalability for real-time or resource-constrained applications.However, the framework is limited to static sign recognition and requires high computational resources, suggesting future extensions toward real-time dynamic gesture recognition, multimodal inputs, and lightweight explainable models for practical deployment. [7]

Chandrasekaran R et al. (2025) presented an interactive sign language learning system designed to support sign language alphabet and word recognition through computer vision and machine learning techniques. The proposed system focuses on providing an engaging and user-friendly learning environment for both beginners and intermediate learners by integrating video demonstrations, interactive quizzes, and real-time gesture recognition. Convolutional Neural Networks (CNN) are employed for recognizing hand gestures corresponding to sign language alphabets, achieving an average accuracy of approximately 90%, while word-level recognition attains around 85% accuracy. The system captures image and video data of hand gestures, performs preprocessing and feature extraction based on hand shape, orientation, and motion, and classifies gestures using trained learning models. Additionally, real-time feedback helps users correct hand positions and movements, improving learning effectiveness. [8]

Arifa Ashrafi et al. (2024) presented a comprehensive study on improving sign language recognition systems using machine learning and artificial intelligence techniques, with a specific focus on Russian and Bengali sign languages. The work addresses both static and dynamic sign recognition by leveraging deep learning architectures such as VGG16, VGG19, and ResNet-50 for image-based classification, and hybrid approaches including CNN–LSTM, Dynamic Time

Warping (DTW), and Hidden Markov Models (HMM) for video-based gesture recognition. Pretrained CNN models are employed for feature extraction, followed by fine-tuning to enhance classification accuracy while preventing overfitting. Experimental results demonstrate that CNN-based models achieve up to 94% accuracy for static sign recognition, with VGG19 outperforming other architectures across both sign languages. However, recognition accuracy for dynamic signs is comparatively lower, ranging between 78% and 88%, highlighting challenges related to temporal variability, frame alignment, and limited annotated video datasets. The study emphasizes the importance of hybrid architectures to capture both spatial and temporal features in dynamic gestures. [9]

Zaw Hein et al. (2021) developed a motion-based sign language recognition system tailored for Myanmar National Sign Language using machine learning techniques. Video input is captured through a webcam and undergoes preprocessing that includes RGB enhancement and transformation to the YCbCr color space for robust skin color detection. Face detection is performed using the Viola–Jones algorithm to assist in accurate hand localization, followed by component detection and localization based on centroid analysis. Leap Motion–based key frame detection is used to identify motion sequences, and horizontal and vertical projection methods are applied for feature extraction. Experimental results demonstrate high recognition accuracy, achieving approximately 94.35% for single-sign groups, 92.32% for double- sign groups, and 85.79% for triple-sign groups across multiple gesture classes. [10]

Luqman et al. (2022) introduced an efficient two-stream deep learning architecture for isolated sign language recognition (ISLR) that captures both spatial and temporal characteristics of sign gestures. The proposed framework consists of a Dynamic Motion Network (DMN) that processes key postures extracted from sign videos using CNN-LSTM models, and an Accumulative Motion Network (AMN) that utilizes a novel Accumulative Video Motion (AVM) technique to encode motion information into a single composite image. Features from both streams are fused and classified using a Sign Recognition Network (SRN). The model was evaluated on multiple datasets including KArSL-190, KArSL-502, and LSA64, achieving superior recognition accuracy and outperforming existing state- of-the-art methods, particularly in signer-independent scenarios. The approach effectively reduces redundant frames and improves recognition of both static and dynamic signs. Despite its strong performance, the system is primarily designed for isolated signs and relies on accurate key posture extraction, which may affect robustness when applied to continuous sign

language recognition tasks.The proposed SignExplainer framework provides visual and quantitative explanations, improving user trust and model interpretability. [11]

Subhangi Kumari et al. (2023) presented a comprehensive review of segmentation and recognition techniques employed for Indian Sign Language (ISL) using machine learning and computer vision approaches. The study discusses the inherent challenges of ISL recognition, including variations in hand gestures, lighting conditions, background complexity, and signer dependency. The authors analyze both traditional hand-crafted feature-based methods such as Histogram of Oriented Gradients (HOG), Local Binary Patterns (LBP), Edge Orientation Histogram (EOH), and modern deep learning techniques including Convolutional Neural Networks (CNNs), Autoencoders, and hybrid learning models. The review also covers various image acquisition devices such as webcams, data gloves, Kinect sensors, and depth cameras, highlighting their advantages and limitations in real-world applications. Comparative analysis of existing literature reveals that deep learning-based approaches significantly outperform conventional machine learning models, achieving recognition accuracies exceeding 90% in controlled environments. The system captures image and video data of hand gestures, performs preprocessing and feature extraction based on hand shape, orientation, and motion, and classifies gestures using trained learning models. Additionally, real-time feedback helps users correct hand positions and movements, improving learning effectiveness. [12]

CHAPTER 3

SYSTEM DESIGN

This chapter describes the overall system design of the proposed Deep Learning and NLP-based real-time sign language recognition system. The design focuses on creating a modular, scalable, and low-latency architecture that enables accurate recognition of Kannada Sign Language gestures and their conversion into meaningful text and speech. The system integrates computer vision, deep learning, natural language processing, and speech synthesis within a web-based framework to provide real-time performance and ease of accessibility.

3.1 SYSTEM ARCHITECTURE

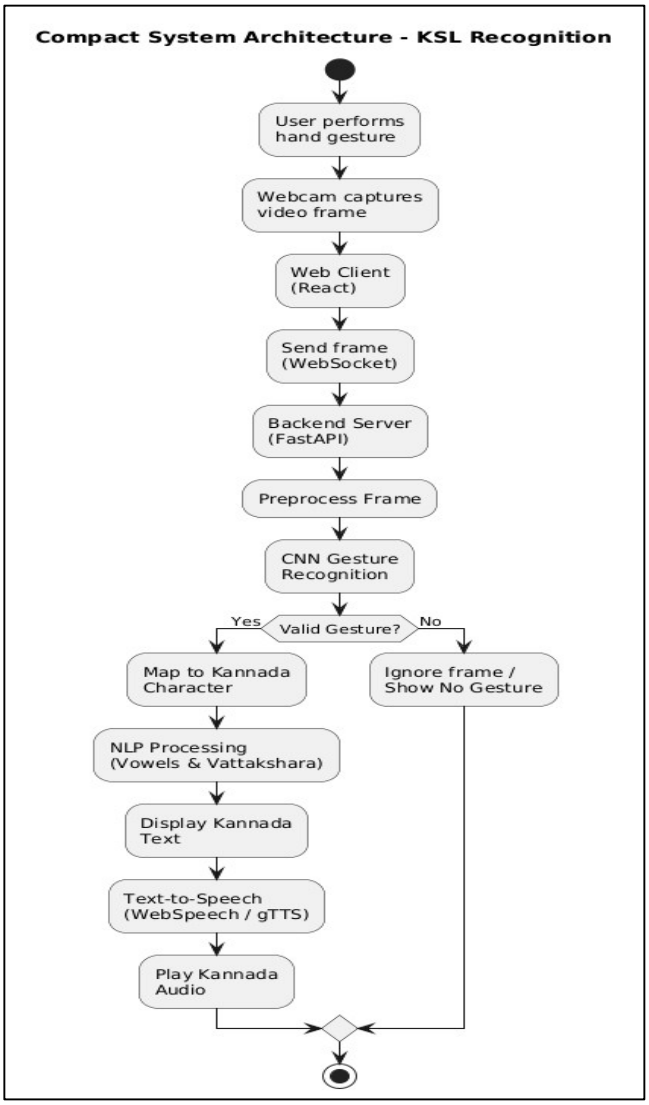

Fig 3.1 System Architecture

The proposed system, as illustrated in the Fig 3.1, follows a layered and modular architecture, where each module is responsible for a specific task in the sign language recognition pipeline. The architecture begins with the input acquisition layer, where live video is captured using a standard webcam through the web browser. This eliminates the need for specialized hardware and ensures easy deployment. The captured video frames are passed to the preprocessing layer, where image normalization, resizing, background noise reduction, and brightness adjustment are performed. These preprocessing steps improve image quality and ensure uniform input to the deep learning model, thereby enhancing recognition accuracy.

The gesture recognition layer forms the core of the system and is implemented using a Convolutional Neural Network (CNN). This layer extracts spatial features such as finger orientation, hand contours, and palm structure from each frame. The trained CNN classifies the input gesture into one of the predefined Kannada Vyanjanagalu classes. The model is optimized to achieve high accuracy while maintaining low inference latency suitable for real-time applications.

The output from the CNN is forwarded to the NLP-based text construction layer. Since Kannada is a context-sensitive language, this layer applies rule-based linguistic processing to generate meaningful text. It handles vowel attachment, vattakshara formation, automatic spacing, duplicate gesture suppression, and confidence-based filtering to ensure grammatically correct Kannada output. The generated text is then passed to the speech synthesis layer, where it is converted into natural Kannada speech. The system primarily uses the browser-based Web Speech API for instant audio output and employs Google Text-to-Speech (gTTS) as a fallback to ensure pronunciation consistency and reliability.

Finally, the web integration layer connects all components through a FastAPI backend and WebSocket communication. The frontend, built using HTML, CSS, JavaScript, and React.js, displays the live video feed, recognized gestures, generated text, and speech output, providing a seamless real-time user experience. The Data flow process is illustrated in Fig 3.2.

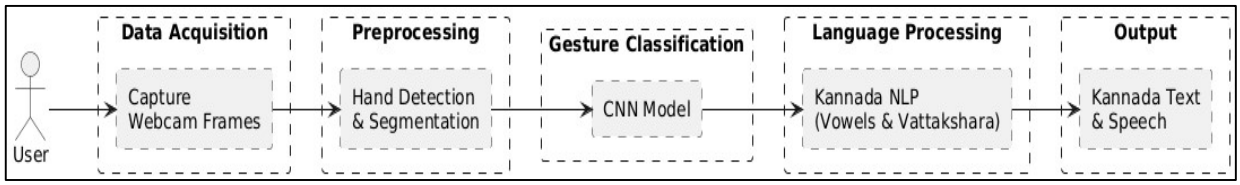

Fig 3.2 Data flow Diagram

3.2 METHODOLOGY

The dataset used for this project consists of images representing Kannada Vyanjanagalu hand gestures. Gesture samples were collected under controlled and semi-controlled environments to include variations in hand orientation, lighting conditions, and backgrounds. Each gesture class was captured multiple times to ensure sufficient data diversity and balanced class distribution. The Use case diagram is illustrated in Fig 3.3.

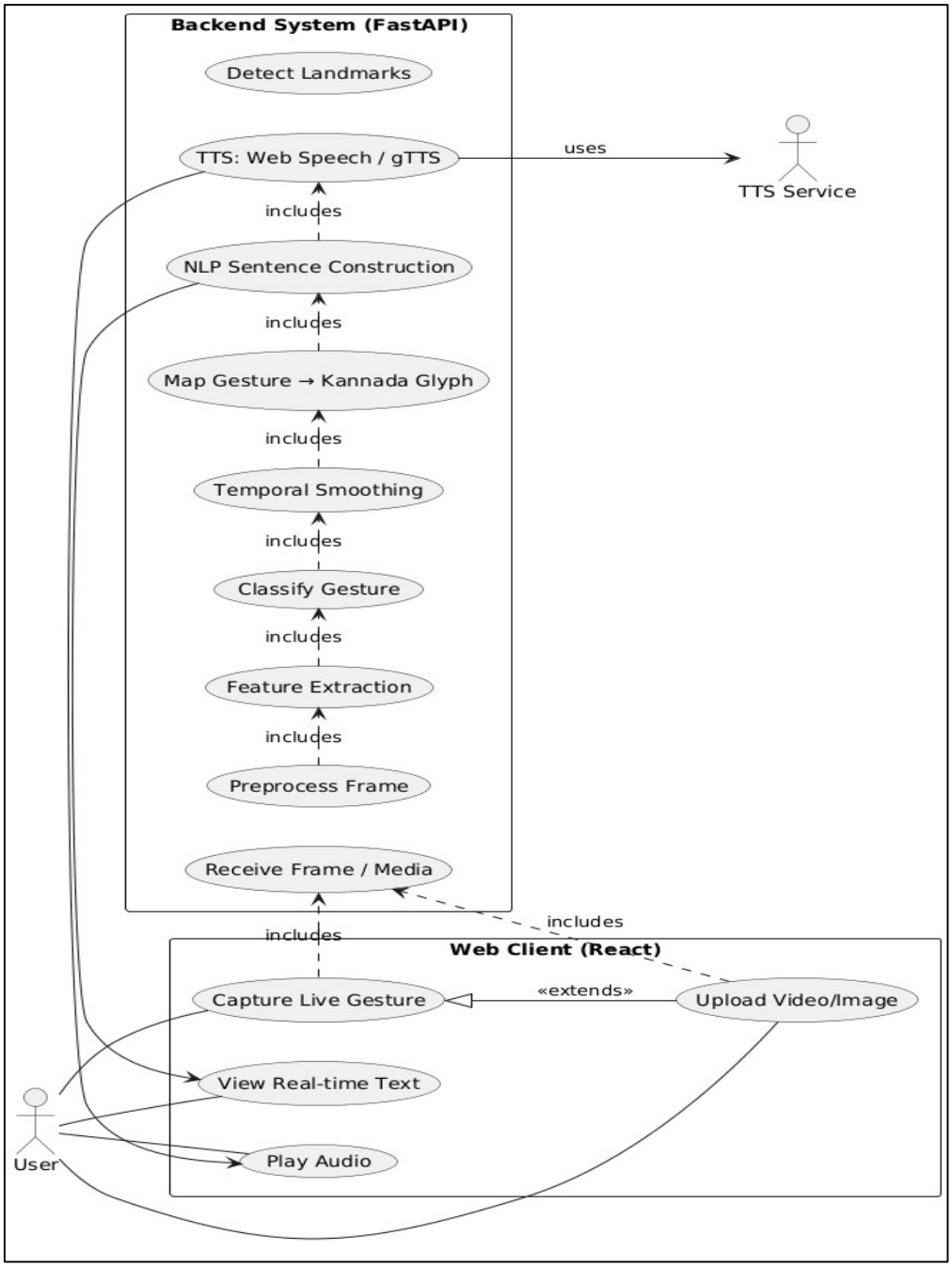

Fig 3.3 Use Case Diagram

- Data preprocessing

  Before training the model, the collected images underwent several preprocessing steps. The images were resized to a fixed resolution suitable for CNN input. Noise reduction and background normalization were applied to minimize environmental interference. Brightness and contrast adjustments were also performed to make the model robust against varying lighting conditions. The preprocessed images were then labeled and organized into class-wise directories.

- Data augmentation

  To improve generalization and prevent overfitting, data augmentation techniques such as rotation, scaling, flipping, and brightness variation were applied. This helped the model learn invariant features and perform reliably under real-world conditions.

- Model training

  A Convolutional Neural Network was designed and trained using TensorFlow and Keras. The dataset was split into training, validation, and testing sets in an 80:10:10 ratio. The CNN learned spatial features of hand gestures through convolution and pooling layers, followed by fully connected layers for classification. The model was trained on a GPU-enabled environment to reduce training time and improve performance.

- Gesture classification

  During real-time execution, each incoming video frame is processed by the trained CNN. The model predicts the most probable gesture class along with a confidence score. Predictions below a predefined confidence threshold are ignored to avoid incorrect outputs.

- NLP-based text construction

  The recognized gesture labels are mapped to Kannada Unicode characters. A rule-based NLP engine processes the character stream to construct meaningful words. This includes vowel attachment, vattakshara formation, removal of repeated characters, and insertion of appropriate spacing. Dictionary-based heuristics are applied for contextual correction.

- Speech synthesis

  The final Kannada text is converted into speech using the Web Speech API. In cases where browser support is limited, gTTS is used as a fallback mechanism. Speech timing, pauses, and pronunciation are optimized to improve clarity and naturalness.

3.3 SEQUENCE DIAGRAM

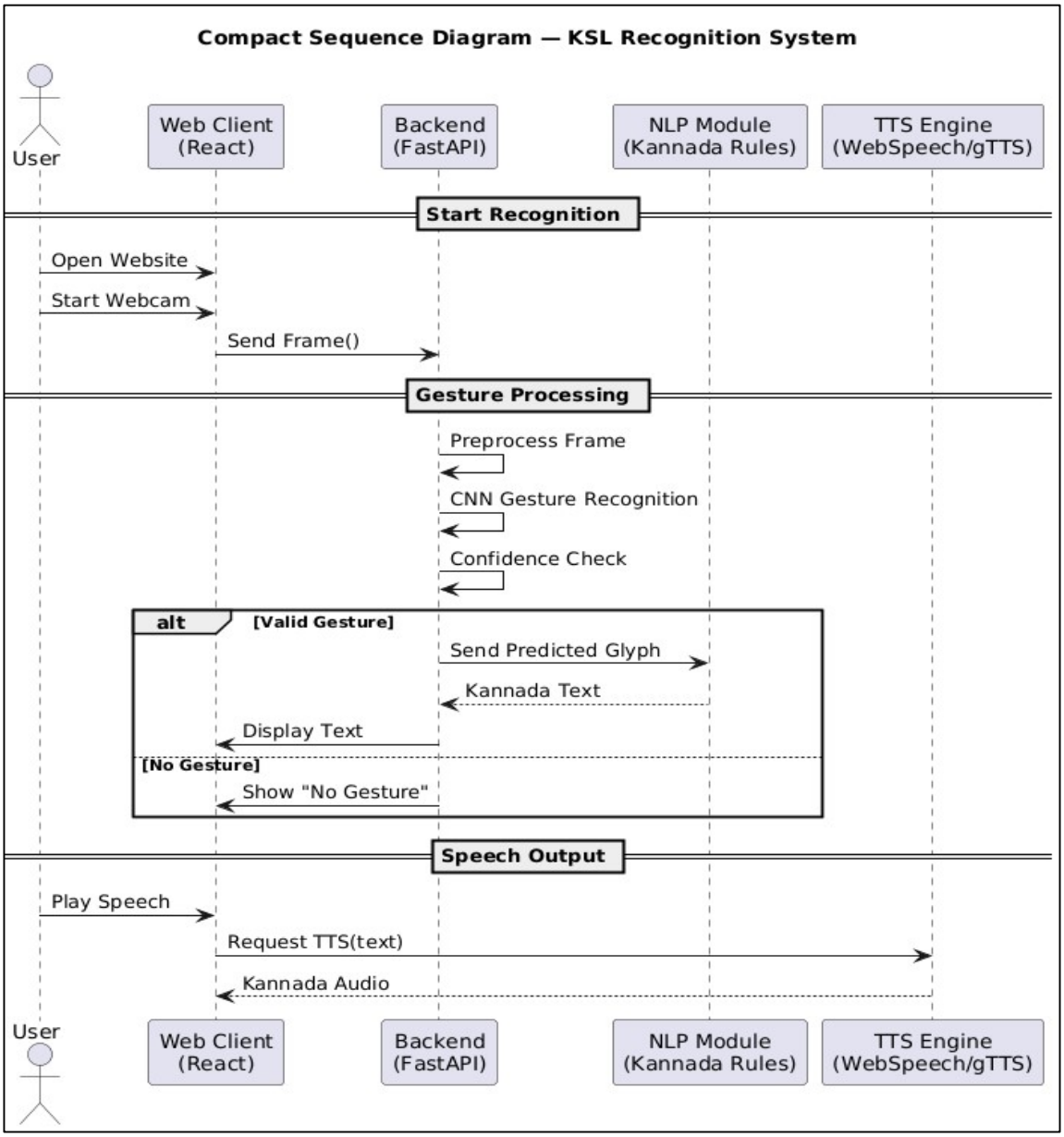

Fig 3.4 Sequential Representation of Working of Project

The sequence of operations in the system begins when the user initiates the application through the web interface as illustrated in Fig 3.4. The webcam captures live video and sends frames to the backend via WebSockets. The preprocessing module processes each frame and forwards it to the CNN model for gesture classification. The predicted gesture label is then sent to the NLP module, which constructs meaningful Kannada text. The generated text is displayed on the user interface and simultaneously passed to the speech synthesis module to produce audio output. This entire sequence repeats continuously, enabling real-time sign-to-text and sign-to-speech conversion.

3.4 CODE

backend/app.py

```
import os, uuid, json
from fastapi import FastAPI, WebSocket, Request, HTTPException, Form
from fastapi.responses import FileResponse, JSONResponse
from fastapi.middleware.cors import CORSMiddleware
from fastapi.staticfiles import StaticFiles
from gtts import gTTS
from model_server import ModelServer
import traceback
MODEL_PATH = "models/kannada_mobilenetv2_best.pth"
IDX_PATH = "models/idx2label.json"
IMG_SIZE = 128
os.makedirs("static/tts", exist_ok=True)
app = FastAPI()
app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"],
    allow_methods=["*"],
    allow_headers=["*"],
)
app.mount("/static", StaticFiles(directory="static"), name="static")
model_server = ModelServer(MODEL_PATH, IDX_PATH, img_size=IMG_SIZE)
@app.get("/health")
async def health():
    return {"status": "ok"}
@app.websocket("/ws/predict")
async def ws_predict(ws: WebSocket):
    await ws.accept()
    try:
        while True:
            data = await ws.receive_json()
            frame = data.get("frame")
```

```
            if not frame:
                await ws.send_json({"error":"no frame"})
                continue
            res = model_server.predict(frame)
            await ws.send_json({"prediction": res["label"], "score": res["score"], "index": res["index"]})
    except Exception:
        # Log the exception server-side so we can see why connections close
        print("WebSocket /ws/predict error:")
        traceback.print_exc()
        try:
            await ws.close()
        except Exception:
            pass
# Accepts JSON body { "text": "..." } or form field 'text'
@app.post("/tts")
async def tts_generate(request: Request):
    """
    Generate TTS using gTTS (fallback). Accepts JSON or form data.
    Returns {"url": "/static/tts/<uuid>.mp3"}
    """
    text = None
    # try JSON
    try:
        data = await request.json()
        if isinstance(data, dict) and "text" in data:
            text = data.get("text")
    except Exception:
        pass
    # try form fallback
    if not text:
        try:
            form = await request.form()
            text = form.get("text")
```

```
    except Exception:
        pass
if not text:
    raise HTTPException(status_code=400, detail="No text provided")
# sanitize short text
text = text.strip()
if len(text) == 0:
    raise HTTPException(status_code=400, detail="Empty text")
uid = str(uuid.uuid4())
filename = f"static/tts/{uid}.mp3"
# generate gTTS (Kannada lang 'kn')
try:
    tts = gTTS(text=text, lang="kn")
    tts.save(filename)
except Exception as e:
    # cleanup possible partial file
    if os.path.exists(filename):
        try: os.remove(filename)
        except: pass
    raise HTTPException(status_code=500, detail=f"TTS generation failed: {e}")
return {"url": f"/{filename}"}
```

backend/model_server.py

```
import io, base64, json
from PIL import Image
import torch
import torch.nn as nn
import torchvision.transforms as T
class ModelServer:
    def __init__(self, model_path, idx2label_path, img_size=128):
        self.device = "cuda" if torch.cuda.is_available() else "cpu"
        self.img_size = img_size
        # Load label mapping
        with open(idx2label_path, "r", encoding="utf-8") as f:
            self.idx2label = json.load(f)
        # Load pretrained model architecture (MobileNetV2)
        from torchvision.models import mobilenet_v2
        self.model = mobilenet_v2(pretrained=False)
        num_ftrs = self.model.classifier[1].in_features
        self.model.classifier[1] = nn.Linear(num_ftrs, len(self.idx2label))
        ckpt = torch.load(model_path, map_location=self.device)
        self.model.load_state_dict(ckpt["model_state_dict"])
        self.model.to(self.device)
        self.model.eval()
        self.transform = T.Compose([
            T.Resize((img_size, img_size)),
            T.ToTensor(),
            T.Normalize(
                mean=[0.485, 0.456, 0.406],
                std=[0.229, 0.224, 0.225]
            )
        ])
    def decode_frame(self, frame_b64):
        if frame_b64.startswith("data:"):
            frame_b64 = frame_b64.split(",")[1]
```

```
    img = Image.open(io.BytesIO(base64.b64decode(frame_b64)))
    return img.convert("RGB")
def predict(self, frame_b64):
    img = self.decode_frame(frame_b64)
    x = self.transform(img).unsqueeze(0).to(self.device)
    with torch.no_grad():
        logits = self.model(x)
        probs = torch.softmax(logits, dim=1)[0]
        idx = probs.argmax().item()
    predicted_label = self.idx2label[str(idx)]
    return {
        "label": predicted_label,
        "score": float(probs[idx]),
        "index": int(idx)
    }
```

CHAPTER 4

IMPLEMENTATION AND TESTING

This chapter describes the implementation details of the proposed deep learning and NLP-based real-time sign language recognition system and presents the testing process used to evaluate its performance. The system integrates gesture recognition, contextual text generation, and speech synthesis into a unified web-based platform. Emphasis is placed on real-time execution, accuracy, and usability under practical conditions.

4.1 HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements

The proposed system is designed to work with minimal and easily available hardware, making it cost-effective and accessible.

The development system requires a standard computer capable of running deep learning frameworks and web services. An Intel Core i3 or i5 processor or higher is sufficient for development and testing purposes. A minimum of 8 GB RAM is required, while 16 GB RAM is recommended for smooth execution during model training and inference. At least 20 GB of free storage space is needed for datasets, trained models, and project files. A GPU is optional but recommended to accelerate the training process.

For real-time operation, the system uses a standard RGB webcam, either built-in or external, to capture hand gesture inputs. No specialized sensors or wearable devices are required. The system can run on commonly available desktop or laptop computers, ensuring ease of deployment.

Software Requirements

The software stack is chosen to support deep learning, computer vision, NLP processing, and web-based deployment. The operating system used is Windows 10 or Windows 11. The core programming language is Python 3.10, which is used for model training, backend development, and integration of different modules. Deep learning and machine learning functionalities are implemented using TensorFlow and Keras. OpenCV is used for video capture, frame preprocessing, and real-time image handling. A rule-based NLP module is implemented using Python to handle Kannada linguistic rules such as vowel attachment and vattakshara formation.

The backend server is developed using FastAPI to enable real-time inference and efficient API handling. WebSocket communication is used to stream video frames from the frontend to the

backend with low latency. The frontend is developed using HTML, CSS, JavaScript, and React.js to provide an interactive user interface. Speech synthesis is implemented using the browser-based Web Speech API, with Google Text-to-Speech (gTTS) used as a fallback for consistent Kannada pronunciation. Development tools such as Visual Studio Code and Google Colab are used for coding, debugging, and model training.

4.2 TESTING

Testing was carried out at multiple levels to ensure the correctness, accuracy, and real-time performance of the system. Both individual module testing and integrated system testing were performed.

- Model Testing

  The trained CNN model was evaluated using a separate test dataset that was not seen during training. Performance metrics such as accuracy, precision, and recall were calculated. The model achieved high classification accuracy for Kannada Vyanjanagalu gestures, demonstrating its ability to generalize well to unseen data. Confidence thresholding was tested to ensure that low-confidence predictions were ignored, reducing false recognitions.
- Preprocessing and Gesture Recognition Testing

  The preprocessing module was tested under different lighting conditions and backgrounds to verify robustness. The system successfully normalized input frames and maintained consistent recognition accuracy even with moderate variations in illumination and hand positioning. Real-time gesture recognition was tested using continuous video input to ensure stable predictions across frames.
- NLP Module Testing

  The rule-based NLP module was tested by feeding sequences of predicted gesture labels. The system correctly performed vowel attachment, vattakshara formation, duplicate suppression, and automatic spacing. Generated Kannada text was verified for grammatical correctness and readability. This testing confirmed that the NLP module effectively converts isolated gesture predictions into meaningful text.
- Speech Synthesis Testing

  The text-to-speech module was tested for clarity, pronunciation, and response time. The Web Speech API provided immediate audio output with low latency. The fallback gTTS mechanism was tested to ensure reliable Kannada speech generation when browser-based synthesis was unavailable. Speech output was found to be clear and understandable.

- Integrated System Testing

  End-to-end testing was performed by running the complete system in real-time using a webcam. The system was tested for continuous gesture input, real-time text display, and simultaneous speech output. Average system latency was observed to be within acceptable limits for real-time interaction. The frontend correctly displayed live video, recognized gestures, generated Kannada text, and played corresponding audio without noticeable delay.

4.3 RESULTS

Dataset Description

The dataset used in this project, "Kannada Vyanjanagalu Sign Language Image Dataset by Prasad Shet" consists of Kannada Sign Language (KSL) hand gesture images representing Kannada Vyanjanagalu (consonants). The dataset was created to support supervised deep learning–based gesture recognition and to address the lack of publicly available datasets for regional sign languages such as Kannada. Each class in the dataset corresponds to a specific Kannada consonant, and multiple gesture samples were collected for every class to ensure balanced representation. The images were captured using a standard RGB camera under controlled and semi-controlled environments. Variations in hand orientation, finger positioning, illumination, and background conditions were intentionally included to improve the robustness of the trained model.

The dataset images were labeled according to the corresponding Kannada Unicode characters and organized into class-wise directories. Prior to training, the dataset was divided into training, validation, and testing subsets in an 80:10:10 ratio. This separation ensured unbiased evaluation of the model's performance on unseen data. To enhance generalization and reduce overfitting, data augmentation techniques such as rotation, scaling, horizontal flipping, and brightness variation were applied.

These augmentations simulate real-world variations and improve the model's ability to recognize gestures under diverse conditions. The prepared dataset enabled effective training of the Convolutional Neural Network (CNN), resulting in high recognition accuracy and reliable real-time gesture classification. The dataset forms a critical foundation for the overall system by ensuring accurate and consistent recognition of Kannada Sign Language gestures.

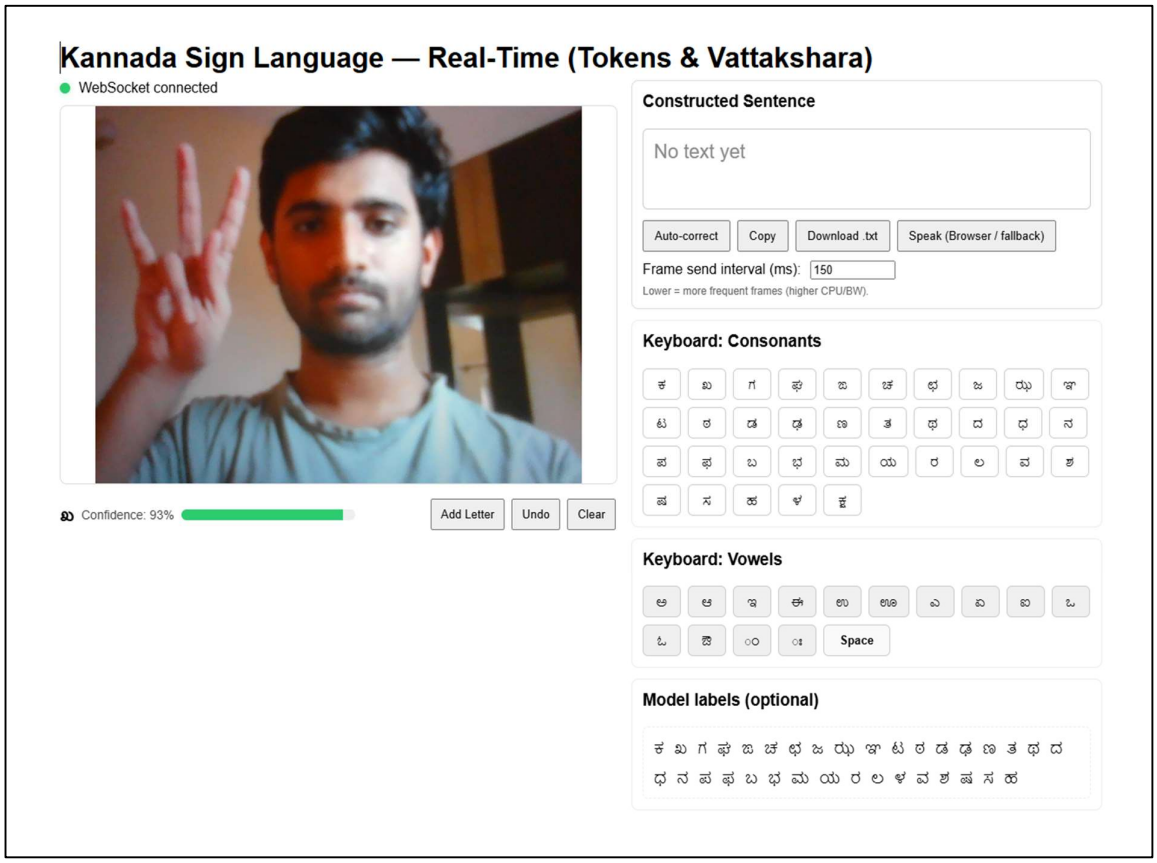

Fig 4.1 Real-Time Kannada Sign Language Gesture Capture Interface

Fig 4.1 illustrates the real-time gesture capture module of the proposed system. Live video is acquired through a webcam and streamed to the backend using WebSocket communication. The interface displays the detected hand gesture along with the predicted Kannada consonant and confidence score. This module serves as the primary input stage for real-time sign language recognition.

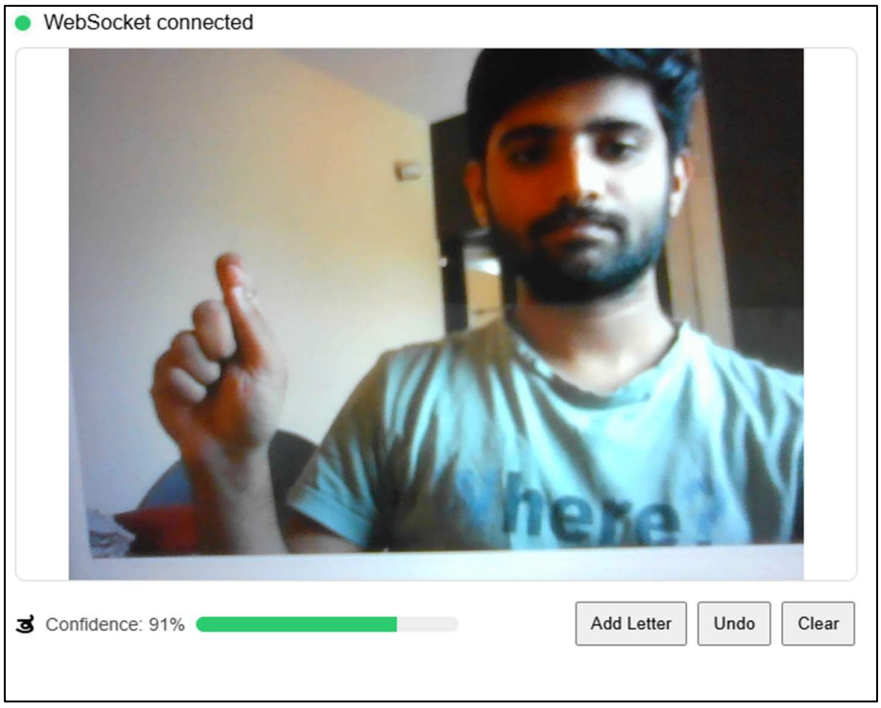

Fig 4.2 CNN-Based Gesture Recognition with Confidence Visualization

Fig 4.2 shows the output of the CNN-based gesture recognition model. The system identifies the performed Kannada sign gesture and displays the corresponding predicted character along with a confidence percentage. The confidence bar indicates the reliability of the prediction, ensuring that only high-confidence gestures are considered for text construction.

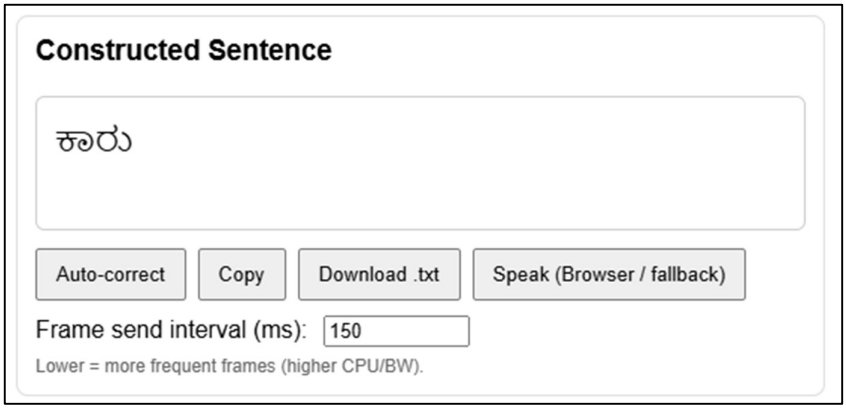

Fig 4.3 Constructed Kannada Text Output Using NLP Rules

Fig 4.3 demonstrates the NLP-based text construction module of the system. Recognized Kannada gesture tokens are combined using rule-based natural language processing techniques to form meaningful Kannada words. The constructed sentence is displayed in real time and can be auto-corrected, copied, downloaded, or converted into speech.

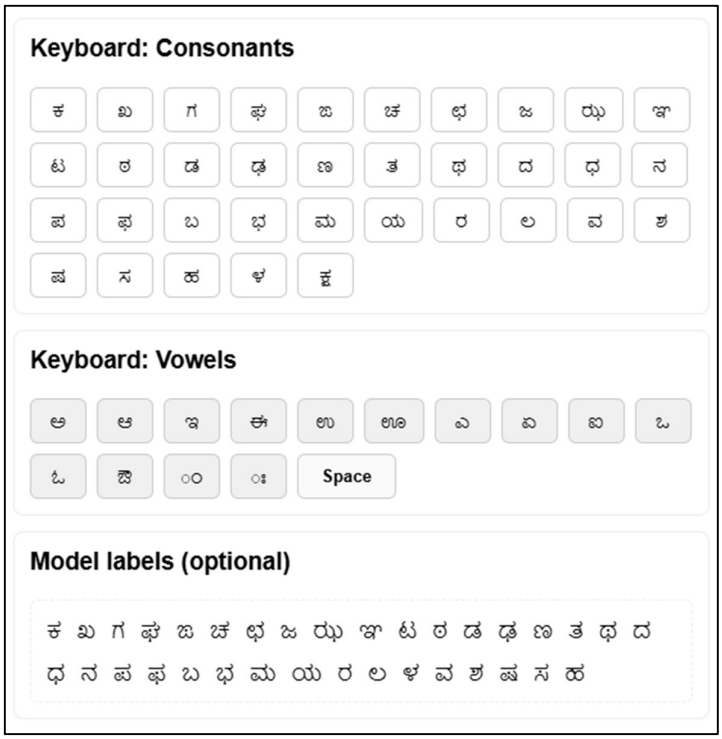

Fig 4.4 Virtual Kannada Keyboard for Consonants and Vowels

Fig 4.4 presents the virtual Kannada keyboard interface integrated into the system. The keyboard includes Kannada consonants (Vyanjanagalu) and vowels (Swaragalu), allowing manual correction or completion of text. This feature enhances usability and ensures accurate sentence formation when automatic recognition requires user assistance.

CHAPTER 5

CONCLUSION

This project successfully designed and implemented a Deep Learning and NLP-based real-time sign language recognition system aimed at improving communication for the Deaf and Hard-of-Hearing (DHH) community. The system focuses on Kannada Sign Language, which is a regionally significant yet technologically underrepresented language. By integrating computer vision, deep learning, natural language processing, and speech synthesis, the project demonstrates an effective end-to-end solution for real-time gesture-to-text and gesture-to-speech conversion. The proposed system utilizes a Convolutional Neural Network (CNN) to accurately recognize Kannada Vyanjanagalu hand gestures from live webcam input. The trained model achieves high recognition accuracy while maintaining low latency, making it suitable for real-time applications. The incorporation of a rule-based NLP module enables the system to generate contextually correct and grammatically meaningful Kannada text, addressing the complexity of Kannada script through vowel attachment and vattakshara formation. Additionally, the integration of a hybrid text-to-speech mechanism ensures clear and natural Kannada speech output. The system is implemented as a web-based application using modern frameworks, allowing easy access without the need for specialized hardware. Experimental evaluation and real-time testing confirmed the reliability, accuracy, and usability of the system under practical conditions. Overall, the project demonstrates the feasibility and effectiveness of applying deep learning and NLP techniques to regional sign language recognition, contributing to inclusive and assistive technologies.

5.1 FUTURE SCOPE

Although the proposed system delivers promising results, there is significant scope for future enhancement and expansion. The current implementation focuses on Kannada consonants (Vyanjanagalu). Future work can extend the system to include Kannada vowels (Swaragalu), compound characters, and complete word-level and sentence-level recognition. The system can be enhanced by incorporating dynamic and continuous gesture recognition using temporal models such as LSTM or Transformer-based architectures. Integration of MediaPipe hand landmark detection can further improve robustness by capturing fine-grained hand movements and reducing dependency on background conditions. Future enhancements may also include mobile application deployment for Android and iOS platforms to improve accessibility. Multilingual support can be introduced to enable translation between Kannada Sign Language and other regional or global

languages. Additional improvements such as user personalization, adaptive learning, and cloud-based model updates can further enhance system performance. Furthermore, the system can be integrated into real-world domains such as education, healthcare, public services, and customer support. With continued dataset expansion and model optimization, the proposed framework can evolve into a comprehensive and scalable sign language translation system that significantly improves communication accessibility for the DHH community.

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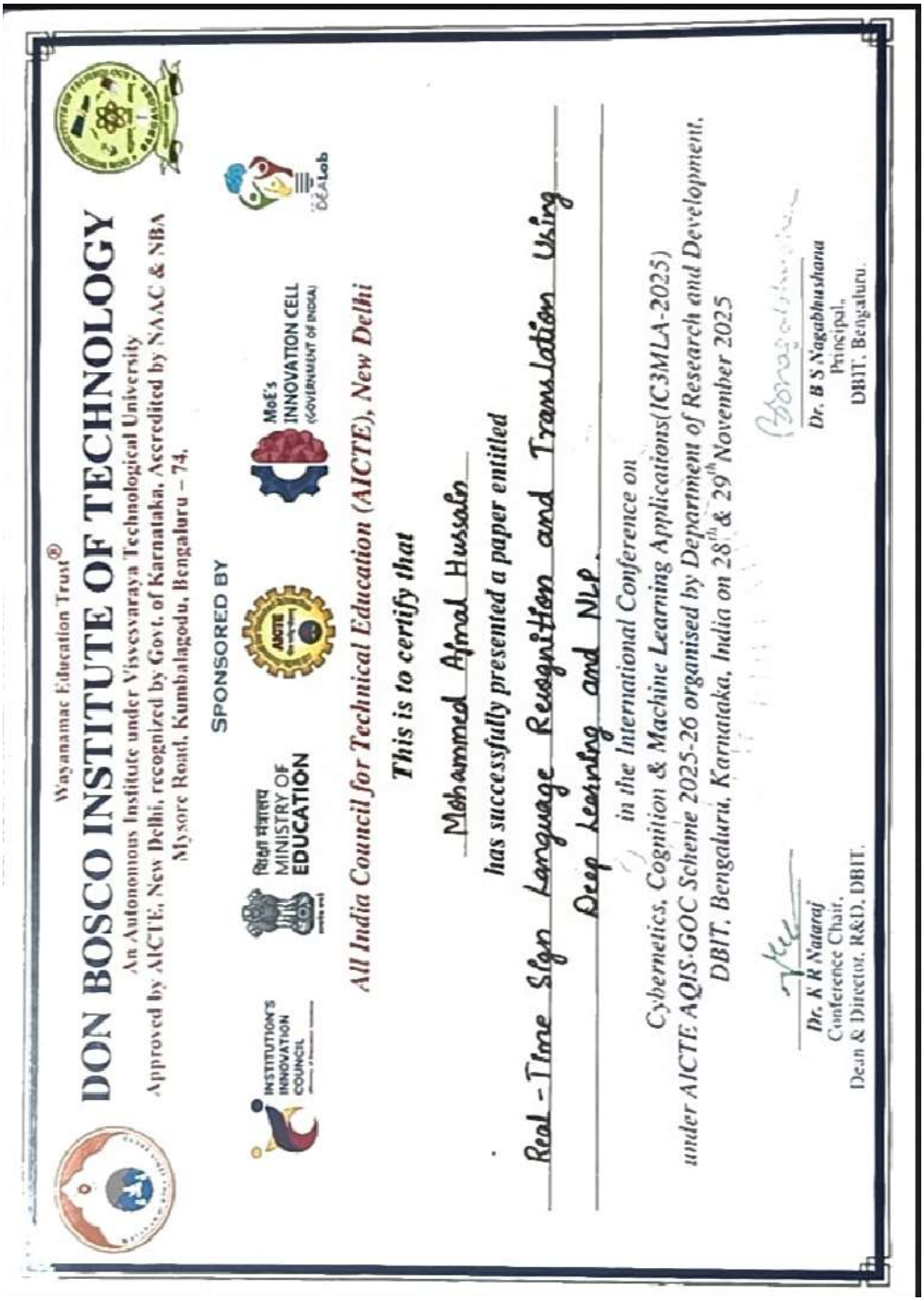

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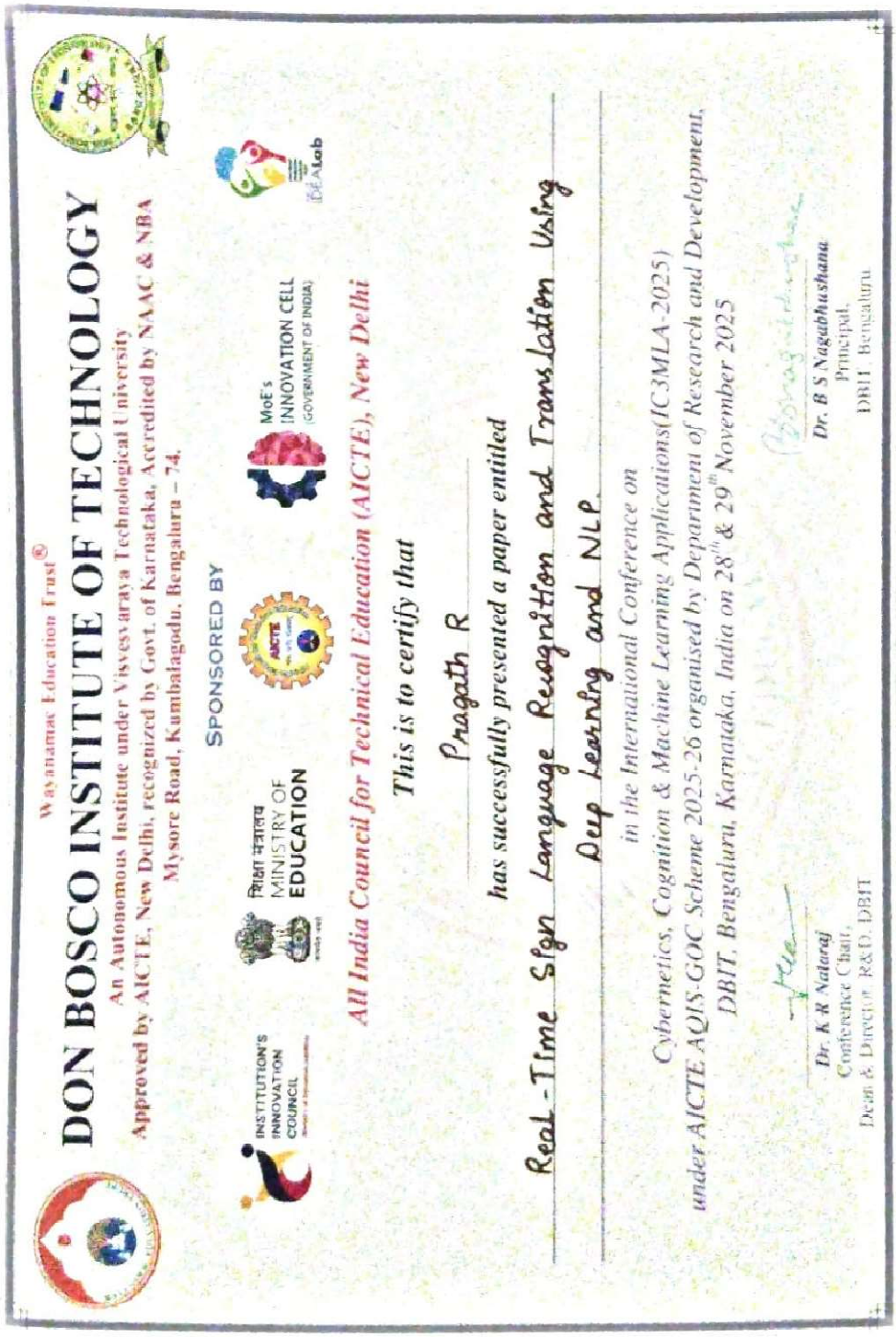

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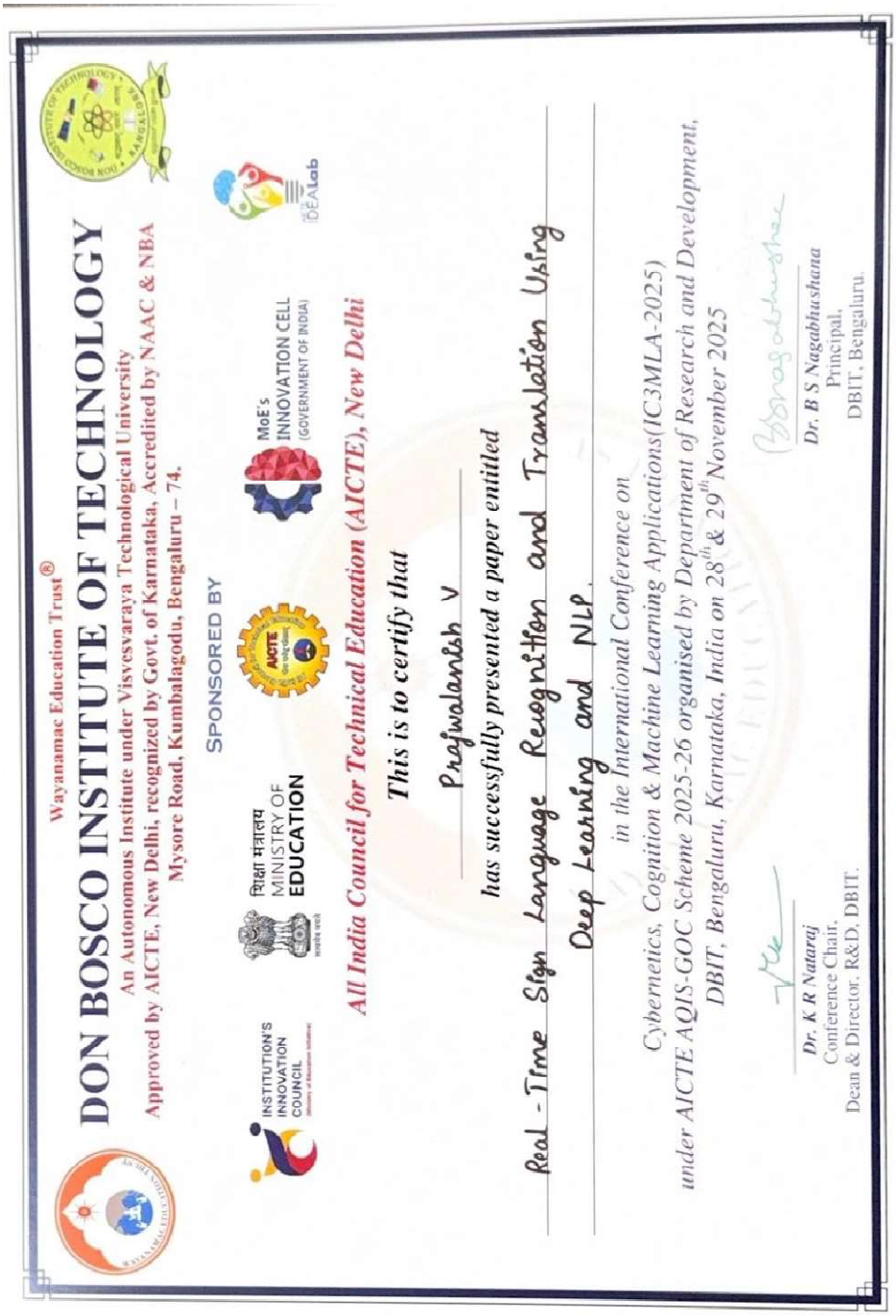

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